Continuous Cardiovascular Risk Estimation via Interpretable Regression with Clinically Derived Features

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***Abstract*—Cardiovascular disease (CVD) remains a major health issue in the global community, and timely intervention and preventive treatment require proper risk evaluation. Clinical risk is continuous and is more effective described in terms of a numerical probability or score even though most of present research forms CVD prediction as a classification problem. In this paper, clinically motivated variables based on patient health record data are utilized to formulate a comprehensible machine learning model to forecast continuous CVD risk. A dataset consisting of 1529 patient samples collected in the Jamalpur Medical College Hospital in Bangladesh is used in the study. The data set consists of 22 clinical and demographic variables and a desired CVD risk score. After pre-processing that included the deletion of redundant features, the solution of missing values, and the transformation of categorical variables into numerical, additional clinically significant variables were generated, including non-HDL cholesterol, mean arterial pressure and pulse pressure, to better represent cardiovascular functioning. Some of the multiple regression models that were evaluated included Ridge (Tuned), Lasso (Tuned), CatBoost, XGBoost, Extra Trees, and ensemble stacking. The Ridge (Tuned) regression model showed that it was significant in modeling the continuous risk score without losing interpretability with the highest level of prediction accuracy at an R2 of 97.12%. To identify relevant clinical predictors of risk, model interpretation was additionally examined with the help of coefficient analysis and SHAP-based feature importance.The paper provides a clear and quantitative approach to cardiovascular risk estimation by positioning CVD risk assessment as a continuous regression problem through the application of clinically grounded feature engineering and interpretable machine learning, offering a transparent framework applicable to resource-limited clinical settings.**

***Index Terms*—cardiovascular disease, risk prediction, ridge regression, explainable AI, SHAP**

I. Introduction

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide and continues to pose a significant global public health challenge. According to the World Health Organization, cardiovascular diseases are responsible for approximately 17.9 million deaths each year, representing nearly one-third of all global deaths [1]. A substantial proportion of these deaths occur in low- and middle-income countries where early screening and preventive healthcare systems may be limited. In South Asian countries such as Bangladesh, the prevalence of cardiovascular disease has increased rapidly due to factors including urbanization, sedentary lifestyles, dietary changes, and the rising incidence of metabolic disorders such as hypertension and diabetes [2]. These trends highlight the urgent need for reliable methods to assess cardiovascular risk at an early stage so that timely preventive interventions can be implemented.

Traditional cardiovascular risk assessment methods, including widely used clinical tools such as the Framingham Risk Score, estimate the probability of developing cardiovascular events based on established risk factors such as age, cholesterol levels, blood pressure, smoking behavior, and diabetes status [3]. While these models have been widely adopted in clinical practice, they rely on simplified statistical assumptions and are often calibrated using specific population cohorts. As a result, their predictive accuracy may vary when applied to different populations or healthcare settings. Moreover, traditional models may not fully capture complex interactions among multiple physiological and behavioral variables that contribute to cardiovascular risk.

Recent advances in machine learning have provided new opportunities to improve disease prediction using data-driven approaches. Machine learning algorithms can analyze large and complex healthcare datasets and identify nonlinear relationships between variables that are difficult to model using conventional statistical techniques [4]. Several studies have applied machine learning methods such as random forests, gradient boosting models, and support vector machines to cardiovascular risk prediction with promising results [5]. These approaches often demonstrate improved predictive performance by learning complex interactions among clinical variables.

Despite these advancements, many existing machine learning studies formulate cardiovascular risk prediction as a bi-

nary classification problem, where patients are categorized as either high-risk or low-risk. While classification models can be useful for screening purposes, they do not fully capture the continuous nature of cardiovascular risk progression. In clinical practice, cardiovascular risk is typically represented as a continuous probability or risk score, reflecting the gradual accumulation of risk factors rather than a strict categorical boundary [6]. Modeling cardiovascular risk as a continuous outcome therefore provides a more nuanced representation of patient health status and allows for more precise estimation of individual risk levels.

Another important consideration in healthcare machine learning is the interpretability of predictive models. Highly complex models may achieve strong predictive accuracy but can be difficult for clinicians to interpret and trust in real-world clinical environments. Interpretable machine learning approaches, particularly regularized regression models such as Ridge and Lasso regression, offer a balance between predictive performance and transparency by clearly illustrating how individual variables influence predicted outcomes [7]. This interpretability is especially important in healthcare settings where understanding the contribution of clinical factors is critical for decision-making.

A key contribution of the present study lies in the incorporation of clinically motivated feature engineering to enhance cardiovascular risk prediction. In addition to conventional clinical measurements, several derived variables were introduced to better represent cardiovascular physiology. These include the total cholesterol to HDL ratio (TC/HDL ratio), non-HDL cholesterol, pulse pressure, and mean arterial pressure. These features provide more informative representations of lipid balance and blood pressure dynamics, both of which are associated with cardiovascular health. By integrating these derived indicators into the modeling framework, the study aims to capture clinically meaningful relationships that may not be fully represented by individual raw measurements alone.

The dataset [8] used in this study originated from Jamalpur Medical College Hospital in Bangladesh. Collected over a twelve-month period ending in early 2025, the repository contains detailed records for 1,529 individual patients. The data encompasses a wide variety of health indicators, including biochemical markers, physical measurements, and lifestyle habits used to evaluate cardiovascular disease risks. Researchers from the Cognitive AI and Informatics Research Lab oversaw the validation of these twenty-two distinct variables to ensure high data quality. Multiple regression-based machine learning models—including Ridge regression, Lasso regression, CatBoost, XGBoost, Extra Trees, and ensemble stacking—were evaluated to determine the most suitable model for predicting the continuous risk score.

By combining clinically informed feature engineering, interpretable regression models, and explainable machine learning techniques, this study proposes a transparent and data-driven framework for cardiovascular risk estimation. The findings may contribute to improved early risk identification and provide insights into the most influential clinical factors associated with cardiovascular disease within the studied population.

II. Literature Review

D'Agostino, Sr. et al. [9] offer a complete recap of the Framingham Risk Functions (FRFs), laying the groundwork for multivariate CVD risk assessment. Their work describes how this evolved from the fairly simple discriminant analysis to more complex Cox proportional hazard survival models fitted over fixed periods (most commonly of 10 years) during which absolute risk can be estimated. Although traditional factors such as age and blood pressure are central, the model's transportability to populations worldwide often needs recalibrating, as they may have baseline risks that vary from those in dominant risk-factor societies. One important concept it highlights is the introduction of "heart age," as a method to make complex risk probabilities more clinically intuitive for patient communication.

Sianga et al. [10] examine the prevalence rate of CVD in Tanzania through a pioneering approach to incorporating geographical factors such as air temperature and humidity into traditional clinical factors. With a data pool of 3,553 cases, the authors used six supervised algorithms to optimize results, proving that the inclusion of environmental and educational factors significantly improved the robustness of the model. The results showed that the XGBoost algorithm was the best approach, yielding an accuracy rate of 95.24% and an AUC-ROC value of 98%, beating traditional regression methods. The authors conclude that regional variations in the environment are major factors that affect the likelihood of CVD and need to be included for early diagnosis, especially in different climates

The issue of overestimation of risk levels in East Asian populations, as described by Western risk prediction models, is addressed by Matheson et al. [11] through the application of Random Survival Forests (RSF) to a cohort of 155,108 Japanese adults. The approach is able to mitigate the non-linear and correlated nature of the annual health check-up data. The authors are able to reduce 169 potential variables to a simple 20-predictor model, yielding an AUC greater than 84% for CAD and 82% for ASCVD. The study highlights the potential for calculating risk trajectories for individuals over time, a tool for prevention in low-risk populations through available screening data

In the study done by Asadi et al. [12], they examined the effectiveness of advanced tree-based models for CVD risk detection in Iran. The study focused on the non-independence of rural areas, where each village is considered a random intercept. The authors compared several models and found that the Generalized Mixed Effect Random Forest (GMERF) performed best with an AUC of 0.80. This method was more effective than using regular decision trees or random forest models in detecting the underlying patterns of non-independence. The study found that age, LDL, and physical activity were the most important factors, again emphasizing the need to address complex local interactions.

In the research article published by Tasnim et al. [13], the authors proposed a new approach to the stacking ensemble technique, particularly designed for low-resource settings in the healthcare domain in Bangladesh, using the same dataset we have used. The authors combined Ridge, Theil-Sen, and Gradient Boosting regression techniques, in which the Ridge regression technique acts as the meta-learner, balancing the stability of the linear model with the flexibility of the nonlinear model. The most interesting aspect of the research article is the use of SHAP (SHapley Additive exPlanations) in the explainable AI domain, which led the model to conclude that the most important factors are BMI and diabetes, achieving an accuracy of 96% and an R^2 value of 0.96, thus proving the usability of the ensemble technique in the healthcare domain, in which structured medical data is scarce

Mettananda et al. [14] constructed the first risk models specific to a human population, targeting Sri Lanka, and proved that regional ML risk models perform much better than global, generic tools. By using longitudinal data from 10 years, the research team was able to construct a risk model using the Random Forest algorithm, where the AUC-ROC was 0.74, while the standard 2019 WHO Southeast Asia risk charts performed poorly with an AUC-ROC of 0.51. The 6-variable parsimonious model was externally validated in a real-world setting, where specificity was 95.0%. This study shows that specific data needs to be considered in order to avoid underestimation of risk in high-risk patients, who are commonly neglected in regional risk calculations

In a study by Bihri et al. [15], an advanced stacking ensemble approach has been presented, which uses a Multilayer Perceptron (MLP) as a meta-learner to process the output from a combination of Random Forest, SVM, and CatBoost models. This stacking ensemble approach aims to detect complex, nonlinear relationships in a combined dataset of 918 records. SMOTE was used to address class imbalance, and SHAP values were used to verify the clinical importance of features such as ST segment slope and cholesterol levels. This MLP-based stacking ensemble approach has resulted in an accuracy of 97.06% and a perfect AUC-ROC score of 1.00.

In the massive-scale research by Xi et al. [16] on 143,043 Chinese hypertensive patients, the authors aimed to enhance short-term risk stratification. By comparing the logistic regression method with the weighted Ensemble model (RF, XGBoost, and Deep Learning average), the authors confirmed the high discriminative power of ensemble methods. The Ensemble model was shown to have an AUROC value of 0.760, far surpassing the standard method. The importance of this research is the inclusion of routine clinical parameters, such as microalbuminuria and carotid thickening, creating an accurate follow-up plan for the largest hypertensive population worldwide.

Hossain et al. [17] employed a cross-sectional study comprising 651 participants in Bangladesh, aiming to find the most dominant predictors of CVD risk, including behavioral and clinical factors. The accuracy of their proposed Random Forest classifier was 98.04%, along with a precision rate of 96.15% and a recall rate of 100%. The SHAP method was employed to find that cholesterol, hypertension, and feelings of hopelessness are critical factors, indicating the connection between mental and physical well-being. The conclusion is that such powerful models will help in better prognosis of patients, especially in countries where there is high mortality from CVDs.

Tawfeek et al. [18] have developed a hybrid ML-AI framework that combines the strengths of both SVM and Particle Swarm Optimization. The framework is used to optimize hyperparameters. The framework was tested using the MIMIC III database and achieved an accuracy of 98.4% and an AUC-ROC of 97.65%. The most interesting feature of the framework is that it has a dynamic weighing system. This enables the framework to learn in real time when patient information, such as genomic or imaging vectors, is missing. SHAP is also used to provide feature-based explanations to factors such as hs-CRP, thus assisting cardiologists in making informed decisions.

The predictive potential of integrating mental health information into a new form of ensemble ML, as described by Dorraki et al. [19], is investigated through a large sample size of 375,145 UK Biobank participants. The study showed that while traditional clinical factors provided a baseline level of accuracy at 71.31%, the inclusion of mental health factors such as depression and anxiety improved the level of accuracy to 85.13% and recall to 89.40%. This study is significant and offers critical proof that mental health assessments are cost-effective, technology-based, and a necessity for a comprehensive patient evaluation for preventive cardiology.

In another study, Jinne et al. [20] examined the effectiveness of supervised ML models using clinical and demographic attributes derived from the CDC to optimize early diagnosis. Out of the supervised ML models examined, the Random Forest classifier performed the best with an accuracy of 84% and an AUC of 92%. The study focused on maximizing sensitivity to reduce false negative cases, especially for high-risk patients without any complications. The feature importance analysis found thalassemia, type of chest pain, and maximum heart rate to be the most important risk factors.

In a study conducted by Xi et al. [21], the authors created a targeted ML risk assessment model for patients with chronic lung diseases (CLDs) due to the fact that the Framingham risk score tends to fail this population of patients. Using a dataset of 1,357 patients, the study was able to create a model using the XGBoost algorithm with an AUC of 78.8% and accuracy of 71.6%. This model uses multidimensional risk factors and identifies BMI, CRP, and grip strength as the top three risk factors. This study emphasizes the importance of disease-specific risk assessment tools due to the unique physiological states of patients with comorbid respiratory conditions.

Hamid et al. [22] presented a fine-tuned predictive model based on a CatBoost algorithm for different stages of heart disease classification, using 918 records from hospitals. The researchers used Rough Set theory to overcome the problems associated with large medical datasets, such as those related

to heart diseases, through attribute reduction and feature selection. The fine-tuned CatBoost model has recorded an outstanding accuracy of 99.02% and an F1 score of 99% compared to other tree-based models. This rapid and accurate detection tool has provided a reliable baseline for early-stage detection, hence reducing CVD-related deaths worldwide.

III. Methodology

This study proposes a machine learning framework for predicting continuous cardiovascular disease (CVD) risk scores using demographic and clinical patient information. The overall workflow of the proposed framework is shown in Fig. 1. The methodology consists of dataset preparation, preprocessing, clinically motivated feature engineering, feature selection, model development, hyperparameter tuning, performance evaluation, and model interpretability analysis. Each stage is described below.

A. Dataset Description

The dataset used in this study was collected from Jamalpur Medical College Hospital, Bangladesh, and contains 1,529 patient records. It includes a set of demographic, behavioral, and clinical variables associated with cardiovascular health. These variables represent a mixture of anthropometric measurements, laboratory test values, blood pressure measurements, and categorical risk indicators. The target variable in this study is the CVD Risk Score, which represents cardiovascular risk as a continuous numerical value. Unlike many previous studies that formulate cardiovascular prediction as a binary classification task, this work treats risk estimation as a regression problem, which allows the model to capture gradual differences in patient risk profiles rather than assigning patients to broad high-risk or low-risk categories.

Figure 2 provides the heatmap of the correlation of the input variables before training the model which gives a detailed picture of the linear relationship between the features and the target variable (CVD risk score). The CVD risk score has a moderate positive association with a number of variables of clinical interest such as BMI (r = 0.61), total cholesterol (r = 0.50), systolic blood pressure (r = 0.47), and non-HDL cholesterol (r = 0.46), which means that an increase in these variables correlates with an increase in cardiovascular risk. Further positive correlations are found with diabetes status (r = 0.45), pulse pressure (r = 0.36), and TC/HDL ratio (r = 0.31) and add to existing clinical relationships. Waist-to-height ratio (r = 0.11) and abdominal circumference (r = 0.08), are anthropometric indicators with weaker, yet positive, correlations with the target variable.

It is also possible to see the high level of inter-feature correlations, which indicates the appearance of multicollinearity in the data, in the heatmap. It is noteworthy that there is a near perfect correlation in height (m) and height (cm) (r = 0.96), whereas in abdominal circumference and waist-to-height ratio, there is a strong association (r = 0.85). Variables relating to lipids are also strongly correlated, total cholesterol and non-HDL cholesterol (r = 0.94), TC/HDL ratio and non-HDL cholesterol (r = 0.82), and HDL cholesterol are negatively correlated with TC/HDL ratio (r = -0.64). Close relationships also exist between blood pressure measures systolic blood pressure and pulse pressure (r = 0.80), mean arterial pressure and diastolic blood pressure (r = 0.80), and systolic and diastolic blood pressure (r = 0.80), pulse pressure and diastolic blood pressure (r = -0.51), respectively.

B. Data Preprocessing

Several preprocessing steps were applied to make the dataset suitable for regression modeling.

1) Clinical field normalization: Some variables required transformation before they could be used for analysis. In particular, the blood pressure field originally stored as a combined text string was separated into its two physiologically meaningful components: Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP). This allowed the model to capture the distinct contribution of each blood pressure component to cardiovascular risk.

2) Categorical variable encoding: Categorical variables were cleaned and encoded into numerical format. String-based variables were first standardized by removing surrounding whitespace and normalizing case formatting. The following mappings were then applied:

- Sex: Male = 1, Female = 0
- Smoking Status: Yes = 1, No = 0
- Diabetes Status: Yes = 1, No = 0
- Family History of CVD: Yes = 1, No = 0

This encoding ensured that clinically meaningful categorical information could be incorporated into the regression models in a consistent manner.

3) Numeric conversion and missing values: All obvious clinical and laboratory variables were converted into numeric format. Non-numeric or inconsistent entries were coerced where necessary. Rows with missing target values were removed, while missing predictor values were handled later within the machine learning pipeline using appropriate imputation strategies.

4) Pipeline-based preprocessing: To avoid data leakage and ensure reproducibility, model preprocessing was embedded inside a pipeline. A ColumnTransformer was used to apply separate preprocessing strategies to numerical and categorical features. Numerical variables were imputed using the median and standardized using StandardScaler, while categorical variables were imputed using the most frequent category and encoded using OneHotEncoder when applicable. This design ensured that preprocessing was learned only from the training data within each fold during cross-validation.

5) Clinically Informed Feature Engineering: A key novelty of this study is the use of clinically grounded feature engineering to enrich the predictive representation of cardiovascular physiology. In addition to the original measured variables, four derived variables were constructed:

1) **Total Cholesterol to HDL Ratio (TC/HDL Ratio)** This ratio was calculated as:

$$\text{TC/HDL Ratio} = \frac{\text{Total Cholesterol}}{\text{HDL}}$$

TABLE I
COMPARISON OF RELATED STUDIES ON CVD RISK PREDICTION

Author	ML Model	CVD Risk Accuracy	Dataset Size	Limitation	Future Direction
Sianga et al. [10]	Optimized XGBoost	95.24%	3,553 cases	Classification-based; imbalanced class distribution in minority groups	Incorporating individual residence types (rural vs. urban)
Matheson et al. [11]	Random Survival Forests (RSF)	AUC $>$ 84%	155,108 adults	Lacks true external dataset for validation	Comparing automated model construction vs. human-led development
Asadi et al. [12]	GMERF (Mixed Effects)	AUC: 80%	9,499 participants	Standard trees showed low specificity (0.59) and binary misclassification	Improving accuracy for non-independent data structures
Tasnim et al. [13]	Stacking (RR + TSR + GBR)	R^2 = 96%	1,529 patients	Classification task; data originates from a single institution	Application of transfer or federated learning frameworks
Mettananda et al. [14]	Random Forest	AUC-ROC: 0.74	2,596 participants	Underrepresented state sector data in training cohort	Development of web/mobile clinical interfaces
Bihri et al. [15]	Stacking (RF + SVM + CatBoost + MLP)	97.06%	918 records	Binary classification limits granularity; risk of overfitting on small data	Real-time clinical deployment and data heterogeneity integration
Xi et al. [16]	Ensemble (RF + XGB + DL)	AUC-ROC: 76%	143,043 patients	Limited number of variables (18) collected at baseline	Adding microalbuminuria to large external cohorts
Hossain et al. [17]	Random Forest	98.04%	651 individuals	Classification focus; performance sensitive to specific attribute selection	Enhancing predictability for associated liver damage
Tawfeek et al. [18]	Hybrid SVM-PSO-AI	98.4%	40,000 (MIMIC-III)	Performance heavily dependent on genomic record accessibility	Adapting framework to other chronic lung conditions
Dorraki et al. [19]	Ensemble (Majority Voting)	85.13%	375,145 participants	Cross-sectional nature cannot establish absolute causal relationships	Randomized prospective studies for clinical workflow deployment
Jinne et al. [20]	Random Forest	84%	14 attributes (CDC)	Baseline models (SVM) were sensitive to feature scaling	Testing on hypothetical profiles for early risk detection
Xi et al. [21]	XGBoost	71.6%, AUC: 78.8%	1,357 CLD patients	Binary outcome; potential omission of environmental and genetic stimuli	Targeted mental health management interventions for CLD
Hamid et al. [22]	Fine-tuned CatBoost	99.02%	918 records	Classification-only model; generalizability across populations requires testing	External validation on independent, diverse datasets

It reflects the balance between overall cholesterol burden and protective high-density lipoprotein cholesterol. This ratio is widely recognized as a lipid-related marker of cardiovascular risk.

2) **Non-HDL Cholesterol**
Non-HDL cholesterol was calculated as:

$$\text{Non-HDL Cholesterol} = \text{Total Cholesterol} - \text{HDL}$$

This feature represents the total amount of cholesterol carried by atherogenic lipoproteins and is clinically relevant for cardiovascular risk estimation.

3) **Pulse Pressure**
Pulse pressure was calculated as:

$$\text{Pulse Pressure} = \text{Systolic BP} - \text{Diastolic BP}$$

Pulse pressure is associated with arterial stiffness and vascular aging and may provide additional information beyond raw blood pressure values.

4) **Mean Arterial Pressure (MAP)**
Mean arterial pressure was calculated as:

$$\text{MAP} = \text{Diastolic BP} \ \frac{\text{Pulse Pressure}}{3}$$

This measure reflects the average arterial pressure across a cardiac cycle and provides a broader representation of hemodynamic status.

These engineered features were added because they capture clinically meaningful relationships that may not be fully represented by the original raw variables alone. Their inclusion was intended to improve model expressiveness while preserving clinical interpretability.

6) Feature Selection and Variable Reduction: To reduce redundancy and simplify the feature space, selected variables were removed such as Estimated LDL (mg/dL) and CVD Risk Level and our target feature was CVD Risk Score prior to model training. The removal process was guided by a combination of redundancy, derivation overlap, and modeling practicality. Variables such as directly duplicated measure-

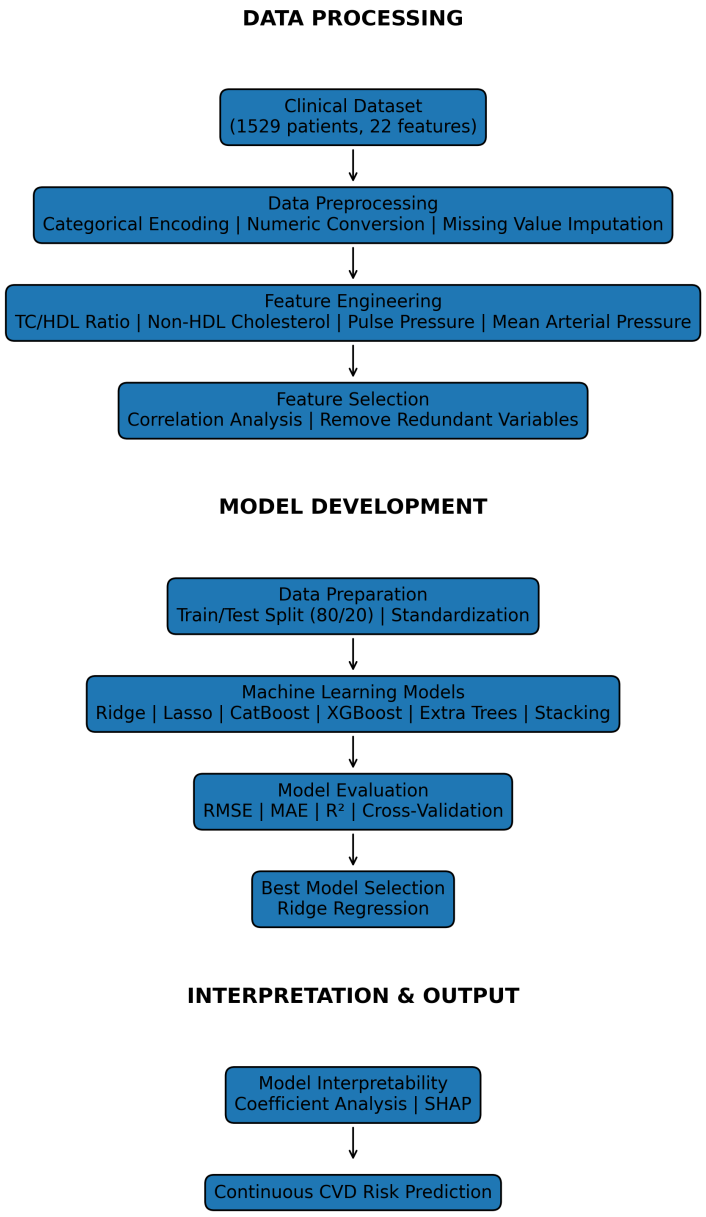


Fig. 1. Block diagram of the proposed framework.

ments, composite fields that had already been decomposed, and certain less informative attributes were excluded where appropriate. This step was used to reduce noise, limit unnecessary multicollinearity,and create a more compact set of predictors while retaining the variables most relevant to continuous CVD risk estimation.

7) Train-Test Splitting and Standardization: After preprocessing and feature engineering, the dataset was divided into training and testing subsets using an 80:20 split, where 80% of the records were used for model training and 20% were reserved for held-out evaluation. Standardization was applied within the preprocessing pipeline using z-score normalization. This was particularly important for regularized linear models such as Ridge and Lasso regression, since these models are sensitive to differences in feature scale. Embedding standardization inside the pipeline ensured that scaling was performed

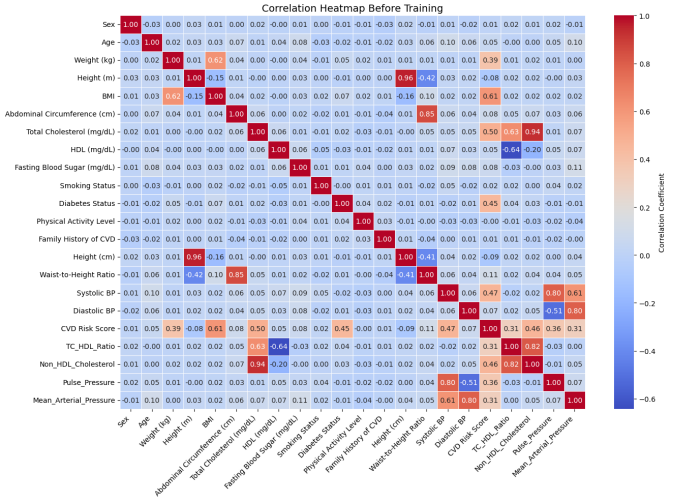


Fig. 2. Correlation heatmap.

consistently and without leakage.

8) Regression Models Evaluated: To identify the most suitable predictive model, six regression approaches were evaluated:

- Tuned Ridge Regression
- Tuned Lasso Regression
- Stacking Regressor
- CatBoost Regressor
- XGBoost Regressor
- Extra Trees Regressor

Ridge and Lasso regression were included as interpretable regularized linear baseline models. Ridge regression applies L2 regularization, which shrinks coefficients and is especially useful when predictors are highly correlated. Lasso regression applies L1 regularization, which can shrink some coefficients to zero, thereby performing implicit feature selection.

CatBoost, XGBoost, and Extra Trees were included as more flexible nonlinear ensemble methods capable of capturing complex interactions among variables. In addition, a Stacking Regressor was implemented to combine the predictions of multiple base learners. The stacking architecture consisted of Lasso, Ridge, Random Forest, and Support Vector Regression (SVR) as base models, with Linear Regression used as the meta-learner.

*9) **Hyperparameter Tuning**:* Because regularization strength strongly affects linear model performance, hyperparameter tuning was performed for both Ridge and Lasso regression.

Tuned Ridge Regression: A pipeline consisting of the preprocessor and a Ridge regressor was optimized using `GridSearchCV`. The regularization parameter α was searched over 20 logarithmically spaced values from 10^{-4} to 10^{4}:

$$\alpha \in \text{logspace}10^{-4}, 10^{4}, 20$$

The optimal value of α was selected using 5-fold cross-validation with RMSE as the optimization criterion.

Tuned Lasso Regression: Lasso regression was tuned in the same way, using the same logarithmic search space for α and the same 5-fold cross-validation strategy.

The tuned Ridge model was later selected as the final model for interpretation because it produced the strongest held-out test performance while preserving interpretability.

10) Model Evaluation: Model performance was evaluated using three standard regression metrics:

- **Root Mean Squared Error (RMSE)**: RMSE measures the average magnitude of prediction error while giving greater weight to larger errors.
- **Mean Absolute Error (MAE)**: MAE measures the average absolute difference between predicted and actual CVD risk scores.
- **Coefficient of Determination** (R^2): R^2 indicates the proportion of variance in the target variable explained by the model.

Performance was assessed in two ways:

- Held-out test set evaluation using the 20% test split
- 5-fold cross-validation to assess model stability and generalization

This dual evaluation approach was adopted to ensure that model selection was not based solely on a single train–test split.

11) Final Model Selection: After comparison of all candidate models, Tuned Ridge Regression was selected as the final model. Although the Stacking Regressor demonstrated slightly stronger cross-validation performance, Tuned Ridge achieved the best performance on the held-out test set, including the lowest test RMSE and the highest test R^2. As the performance differences among the leading models were relatively small, final model selection was based not only on predictive accuracy but also on interpretability, reproducibility, and clinical transparency.

Accordingly, Tuned Ridge Regression was considered the most suitable final model for continuous CVD risk estimation in the present study.

Model Interpretability: To explain the predictions of the final model, interpretability analysis was performed using two complementary approaches:

Coefficient-based Interpretation: Because Ridge regression is a linear model, its coefficients provide direct information about the direction and relative magnitude of predictor effects after preprocessing and scaling.

SHAP Analysis: To obtain a more detailed model-level explanation, SHAP (SHapley Additive exPlanations) analysis was performed on the final Tuned Ridge model. SHAP values quantify the contribution of each feature to the predicted risk score, both at the global level and for individual observations.

Several SHAP visualizations were generated, including:

- a global SHAP feature importance bar plot,
- a SHAP summary plot,
- a feature importance ranking table,
- dependence plots for the most influential features, and
- an individual waterfall explanation.

These analyses were used to identify the most influential predictors of continuous CVD risk and to assess whether the patterns learned by the final model were clinically meaningful.

IV. Results

A. Model Evaluation

Table II presents the predictive performance of all evaluated regression models on the test set and under 5-fold cross-validation. Among the candidate models, Tuned Ridge, Stacking, and Tuned Lasso clearly outperform CatBoost, XGBoost, and Extra Trees. Following the paper structure guideline, the results are summarized through tables, figures, and supporting text, with additional analysis through explainable AI for the final model.

Tuned Ridge was the best performing on the held-out test set, with an overall performance of Test RMSE of 0.4171, Test MAE of 0.1814, and Test R2 of 0.9712. Next was the Stacking regressor with a Test RMSE of 0.4186 and Test R2 of 0.9709 and a Tuned Lasso Test RMSE of 0.4220 and Test R2 of 0.9705. CatBoost, XGBoost, and Extra Trees, in contrast, had larger prediction errors. The Tuned Ridge model achieved the best results in terms of R2 (0.9712) and MAE, and outperformed non-linear ensemble approaches such as XGBoost. This is because the model can cope with the high correlation of features extracted from clinical data. Ensemble models may capture complex interactions, but the Ridge model's regularization was more effective in stabilizing predictions across the clinical cohort, particularly for blood pressure-related variables

The cross-validation performance was also similar. The best average cross-validation results were obtained with Stacking, a CV RMSE of 0.4317 (±0.0236) and a CV R2 of 0.9682 (±0.0039), and then Tuned Lasso and Tuned Ridge. The difference in the performance of the top three models was quite small even though Stacking had a slight lead in the cross-validation. We have chosen to use Tuned Ridge regression as the final model based on both predictive accuracy and interpretability since it had the highest held-out test performance, and it is simpler and transparent compared to the ensemble-based Stacking method.

B. Regression Performance and Error Analysis

Figure 3 illustrates that the risk score of cardiovascular disease (CVD) as estimated by the Ridge (Tuned) regression model matches the real risk scores. The scatter plot indicates that the majority of the points are near to the diagonal line. This trend indicates that the model is capable of demonstrating the relationship between the predictor variables and the CVD risk score well. All the risk values are very close along the diagonal, which means that the model provides correct predictions both to low- and high-risk individuals. There are only a few observations that cannot be fitted on the line and they are mostly midway along the risk spectrum. This distribution explains why the regularization effect of Ridge (Tuned) regression ensures that the model remains stable and is still able to make good predictions. The fact that the predicted

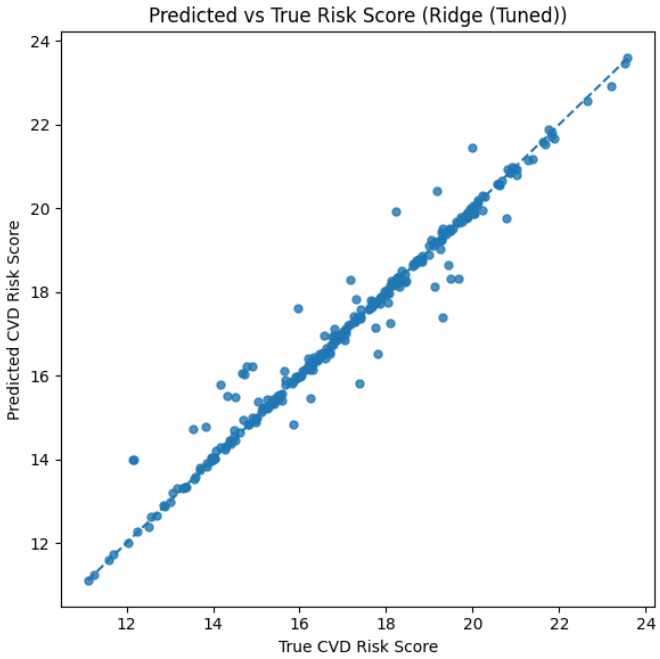


Fig. 3. Predicted VS True Risk Score(Ridge(Tuned))

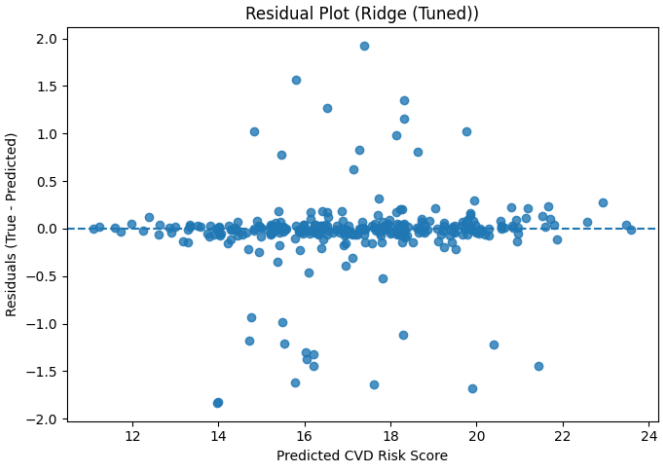


Fig. 4. Residual Plot (Ridge(Tuned))

and actual scores are very close demonstrates that the model is capable of making correct forecasts.

Figure 4 contains the residual plot of the Ridge (Tuned) regression model. The difference between the true and the predicted CVD risk score or the residuals, is plotted against the predicted values. The majority of the residual values are near the horizontal zero line indicating that the model prediction in general is often near to the actual values and that the total error of prediction is also minimal. The residual values appear to be dispersed randomly over this reference line rather than taking a consistent pattern implying that the model does not exhibit a serious systematic bias over the entire range of the scores of the predicted risks. That is, the model does not necessarily

TABLE II
PERFORMANCE COMPARISON OF REGRESSION MODELS

Model	Test_MSE	Test_RMSE	Test_MAE	Test_R2	CV_RMSE_Mean	CV_RMSE_STD	CV_MAE_Mean	CV_MAE_STD	CV_R2_Mean	CV_R2_STD
Ridge (Tuned)	0.1739	0.4171	0.1814	0.9712	0.4369	0.0184	0.1896	0.0062	0.9675	0.0032
Stacking	0.1753	0.4186	0.1657	0.9709	0.4317	0.0236	0.1660	0.0043	0.9682	0.0039
Lasso (Tuned)	0.1781	0.4220	0.1692	0.9705	0.4305	0.0257	0.1663	0.0070	0.9683	0.0042
CatBoost	0.2327	0.4824	0.2864	0.9614	0.4917	0.0270	0.2906	0.0100	0.9587	0.0048
XGBoost	0.2557	0.5056	0.3191	0.9576	0.5160	0.0252	0.3352	0.0100	0.9545	0.0050
Extra Trees	0.3378	0.5812	0.4033	0.9440	0.6029	0.0248	0.4172	0.0192	0.9381	0.0055

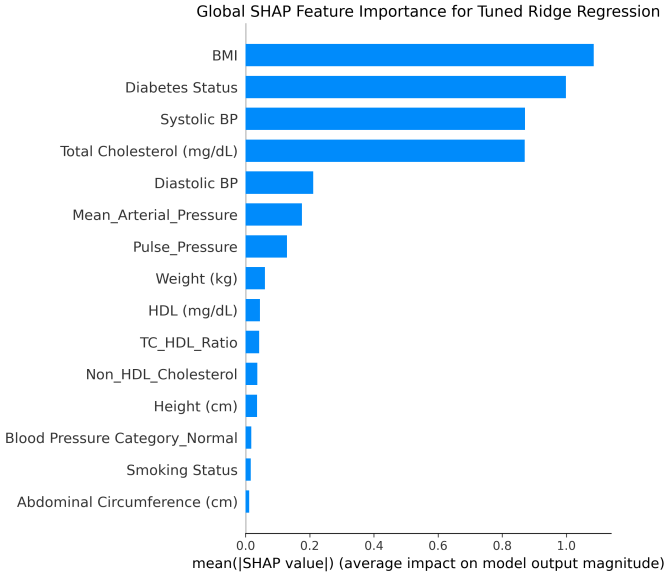


Fig. 5. Global SHAP Feature Importance for Tuned Ridge Regression.

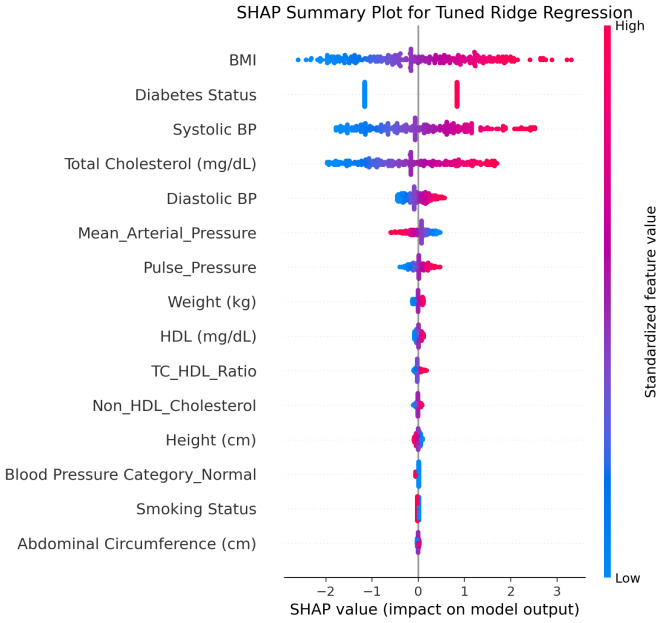


Fig. 6. SHAP Summary Plot for Tuned Ridge Regression.

over- or underestimate the CVD risk of some value ranges. There are some observations in which the residuals are large (positive or negative), but these are infrequent. The general pattern of residuals indicates that the Ridge regression model is relatively stable and has fairly correct predictions and the errors remain mostly close to zero throughout the spectrum of the predicted risks.

C. SHAP Analysis

Figure 5 shows the final model. The SHAP importance plot of the world revealed that the most significant predictors of the continuous CVD risk score were BMI, Diabetes Status, Systolic BP, and Total Cholesterol (mg/dL). In Figure 6, the SHAP summary plot also indicated that the greater values of BMI, systolic blood pressure, and total cholesterol were, the more they could predict the risk, and diabetes status also demonstrated a strong positive contribution to the model output. These results are consistent clinically and indicate that the final model is based on meaningful cardiovascular risk factors.

The engineered variables presented in this study such as TC/HDL Ratio, Non-HDL Cholesterol, Pulse Pressure, and Mean Arterial Pressure had lower contributions of SHAP than the overall leading clinical predictors. Nevertheless, these features still offered complementary physiological data and helped to construct the whole predictive system. On the whole, the findings indicate that a continuous cardiovascular risk estimation may be effectively conducted with the help of interpretable regression-based machine learning, with Tuned Ridge providing the best trade-off between the predictive accuracy and model transparency.

V. LIMITATIONS

Data from one institution. This study utilized a dataset exclusively sourced from Jamalpur Medical College Hospital, Bangladesh, consisting of 1,529 patient records. The generalizability of the findings to larger populations is restricted by their single-site origin. Variations in patient demographics, clinical practices, and referral patterns among institutions may not be adequately represented in this sample. There was no external validation on a separate dataset from a different institution or geographic area, and it is still unclear how well the proposed model works with other populations.

Dependence on clinically derived risk scores. The target variable in this study is the CVD Risk Score , which is a composite metric based on clinical assessment at Jamalpur Medical College Hospital, rather than longitudinal outcomes such as confirmed cardiac events or death. The model essentially serves as a transparent digital twin of the clinical

scoring process, while the score itself is a validated clinical judgment at the point of care. This probably explains the better performance of the regularized linear models Tuned Ridge compared to the complex nonlinear ensembles, since the underlying clinical logic is inherently structured around established physiological correlations. Additional studies with long-term follow-up of patients are needed to confirm these results and to relate the predicted risk scores to actual cardiovascular events.

Multicollinearity among predictors. The correlation heatmap shows that there are strong correlations between features in the dataset, especially between blood pressure components, lipid-derived variables, and anthropometric measurements. Even though Tuned Ridge regression is meant to deal with correlated predictors using L2 regularization, the high level of multicollinearity makes it hard to understand what each coefficient and SHAP value means. Assigning predictive significance to individual features necessitates caution, as correlated features may share or redistribute contributions among themselves instead of indicating independent clinical effects.

VI. Conclusion

This study presents an interpretable machine learning framework for predicting continuous cardiovascular disease (CVD) risk scores using clinical and demographic data. By formulating CVD risk estimation as a regression problem rather than a classification task, the proposed approach enables more precise and individualized risk assessment.

This work additionally incorporated clinically derived features including TC/HDL ratio, non-HDL cholesterol, pulse pressure, and mean arterial pressure to enrich the physiological representation of the input space. These features were designed to better represent cardiovascular physiology and enhance the predictive capability of the models.

Multiple regression algorithms were evaluated, including regularized linear models and ensemble-based approaches. Among these, Tuned Ridge regression achieved the best overall performance on the held-out test set, with high predictive accuracy and strong generalization. Although the Stacking model showed slightly improved cross-validation performance, the differences between the top models were relatively small. Therefore, Tuned Ridge was selected as the final model due to its superior balance between accuracy, interpretability, and reproducibility.

Model interpretability analysis using SHAP demonstrated that the final model is primarily driven by clinically meaningful predictors, particularly BMI, diabetes status, systolic blood pressure, and total cholesterol. These findings are consistent with established medical knowledge and indicate that the model captures realistic cardiovascular risk patterns rather than relying on spurious correlations. Overall, the results suggest that continuous CVD risk estimation can be effectively achieved using interpretable regression-based machine learning. The proposed framework provides a transparent and clinically relevant approach for cardiovascular risk prediction, which could support early detection and preventive healthcare strategies.

Future work may focus on incorporating multi-center datasets, expanding feature sets to include lifestyle and genetic factors, and exploring longitudinal modeling approaches to better capture temporal changes in cardiovascular risk. Additionally, hybrid models that combine interpretability with controlled nonlinearity may further enhance predictive performance while maintaining clinical usability.

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